参考答案

一、

1.T；2.T；3.T；4.F；5.F；6.F；7.A；8.A；9.B；10.B；11.A；12.A；13.C；14.C；15. $\Delta_{g\ 物\ f}$, Δ_{g}；

16.> < = =; 17. 压缩 冷凝冷却 节流 蒸发

二、

解：根据热力学第一定律Δ s (因为$W_s=0$)，则

$$Q=\Delta H=H_2-H_1=3117-3213=-96(\mathrm{kJ\cdot kg^{-1}})$$

$$W_L=T_0\ g\ 0\ f\ 0\ 2\ 1\ \frac{Q}{T_0}$$

$$=298\times\left[6.621-6.7690-\left(\frac{-96}{298}\right)\right]=52.045(\mathrm{kJ\cdot kg^{-1}\cdot K^{-1}})$$

三、

解：

$$W_{id}=G(\Delta H-T_0\Delta S)=Gc_P[(T_2-T_1)-T_0\ln\frac{T_2}{T_1}]$$

$$=14400\times3.4\times[(278-298)-298\times\ln\frac{278}{298}]$$

$$=34407(\mathrm{kJ\cdot kg^{-1}})$$

已知

$\varepsilon=\frac{Q_0}{W_S}=4$，$Q_0=Gc_P(T_1\text{-}T_2)=14400\times3.4\times(298-278)=979200(\mathrm{kJ})$，则

$$W_S=\frac{Q_0}{\varepsilon}=\frac{979200}{4}=244800(\mathrm{kJ\cdot h^{-1}})$$

$$W_L=W_S\text{-}W_{id}=244800-34407=210393(\mathrm{kJ\cdot h^{-1}})$$

四、

解：在恒沸点满足：$P_{1\ 1}^{s'}\ {}_{2}^{s'}\ {}_{2}$，则 $\ln\frac{P_1^S}{P_2^S}=\ln\frac{\ 2}{\ 1}=B(x_1^2\text{-}x_2^2)$

将恒沸点组成：$x_1^{az}=0.294$，$x_2^{az}=0.706$与P_1^S，P_2^S代入上式得：B=-2.1，因此，

$\ln\gamma_1=-2.1x_2^2$，$\ln\gamma_2=-2.1x_1^2$

当$x_1=0.5$时，计算可得：$\gamma_1=0.592$，$\gamma_2=0.592$

$$p=p_1^S\cdot x_1\gamma_1+p_2^S\cdot x_2\gamma_2=(75.20+3.66)\times0.5\times0.592=31.61(\mathrm{kPa})$$

$$y_1=\frac{p_1^S x_1\gamma_1}{p}=\frac{75.2\times0.5\times0.592}{31.61}=0.7042$$

五、

$$\frac{G^E}{RT} = x_1 ln\gamma_1 + x_2 ln\gamma_2 = x_1 ln\frac{f_1'}{f_1 x_1} + x_2 ln\frac{f_2'}{f_2 x_2} = x_1 ln\frac{f_1'}{x_1} + x_2 ln\frac{f_2}{x_2} - x_1 \ln f_1 - x_2 \ln f_2$$

$$= \ln f - x_1 \ln f_1 - x_2 \ln f_2$$

上式中：$\ln f_1 = \lim_{x_1 \to 1} \ln f = -\lim_{x_2 \to 1}(C + Dx_1) = C + D$

$$\ln f_2 = \lim_{x_1 \to 0} \ln f = -\lim_{x_1 \to 0}(C + Dx_1) = C$$

将其代入式（1）整理得：$\frac{G^E}{RT}$=0，所以必有，$ln\gamma_1$=0，$ln\gamma_2$=0，溶液为理想溶液。

六、

因 $ln\gamma_1 = \left[\frac{\partial\left(nG^E/RT\right)}{\partial n_1}\right]_{T,P,n_2}$，则 $d\left(nG^E/RT\right) = \left(ln\gamma_1\right)dn_1$

$$d\left(nG^E/RT\right) = \frac{a}{\left(1+\frac{bx_1}{x_2}\right)^2}dn_1$$

$$= \frac{a}{\left(1+\frac{bn_1}{n_2}\right)^2}dn_1 = \frac{a\cdot\frac{n_2}{b}}{\left(1+\frac{bn_1}{n_2}\right)^2}d\left(1+\frac{bn_1}{n_2}\right)$$

积分得：

$$\int_0^{nG^E/RT} d(nG^E/RT) = \int_0^{n_1} \frac{a\cdot\frac{n_2}{b}}{\left(1+\frac{bn_1}{n_2}\right)^2}d\left(1+\frac{bn_1}{n_2}\right) = \frac{a\cdot\frac{n_2}{b}}{\left(1+\frac{bn_1}{n_2}\right)^2}\Big|_0^{n_1}$$

$$nG^E/RT = -\left[\frac{n_2\cdot\frac{a}{b}}{\left(1+\frac{bn_1}{n_2}\right)} - n_2\cdot\frac{a}{b}\right] = \frac{n_1\cdot a}{\left(1+\frac{bn_1}{n_2}\right)}$$，即：$\frac{G^E}{RT} = \frac{ax_1x_2}{x_2+bx_1}$